

Multiple Choice (10 marks)

1. The heart does not extract oxygen and nutrients from the blood within the atria and ventricles. How is the heart supplied with its metabolic needs?
 - (a) The heart synthesizes its own nutrients internally without reliance on external sources.
 - (b) The heart relies on neural control for its metabolic needs.
 - (c) The heart extracts nutrients from the atria and ventricles.
 - (d) The heart's metabolic needs are met by the lymphatic system.
 - (e) The heart depends on its own blood supply from the coronary vessels.
2. Hardy-Weinberg equilibrium generally assumes all of the following EXCEPT
 - (a) a large population
 - (b) genetic drift
 - (c) random mating
 - (d) absence of selection
3. Why is the drop in blood pressure important in retaining fluid within the capillaries?
 - (a) The drop in blood pressure causes blood clotting
 - (b) The drop in blood pressure causes an increase in plasma protein concentration
 - (c) The drop in blood pressure stimulates the excretion of fluids by the kidneys
 - (d) The drop in blood pressure causes dehydration
 - (e) The drop in blood pressure triggers the release of antidiuretic hormone
4. Two related species evolve in a way that makes them less similar.
 - (a) Coevolution
 - (b) Punctuated equilibrium
 - (c) Directional selection
 - (d) Divergent evolution
 - (e) Convergent evolution
5. A plant grows in the opposite direction of the gravitational force. This is an example of
 - (a) positive chemotropism
 - (b) positive phototropism

- (c) negative phototropism
- (d) negative chemotropism
- (e) positive hydrotropism

True / False (10 marks)

Answer True or False

6. True or False: Both robins and dogs are warm-blooded animals that maintain a constant body temperature regardless of environmental changes.
7. True or False: Synthesis of an RNA/DNA hybrid can occur with RNase and a DNA primer without the need for reverse transcriptase.
8. True or False: Coevolution refers to the reciprocal evolutionary changes that occur between closely related species, such as in predator-prey dynamics.
9. True or False: The wing of a butterfly is homologous to the wing of a bird, as both serve the function of flight.
10. True or False: Centrosomes play a crucial role in regulating the checkpoints of cell division.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Discuss the stages of ontogeny in a human being from zygote to mature fetus, analyzing the significance of these developmental processes for overall human growth and health.
12. Discuss the fate of a vesicle formed through phagocytosis, including its implications for cellular processes and potential interactions within the cytoplasm.

End of Paper